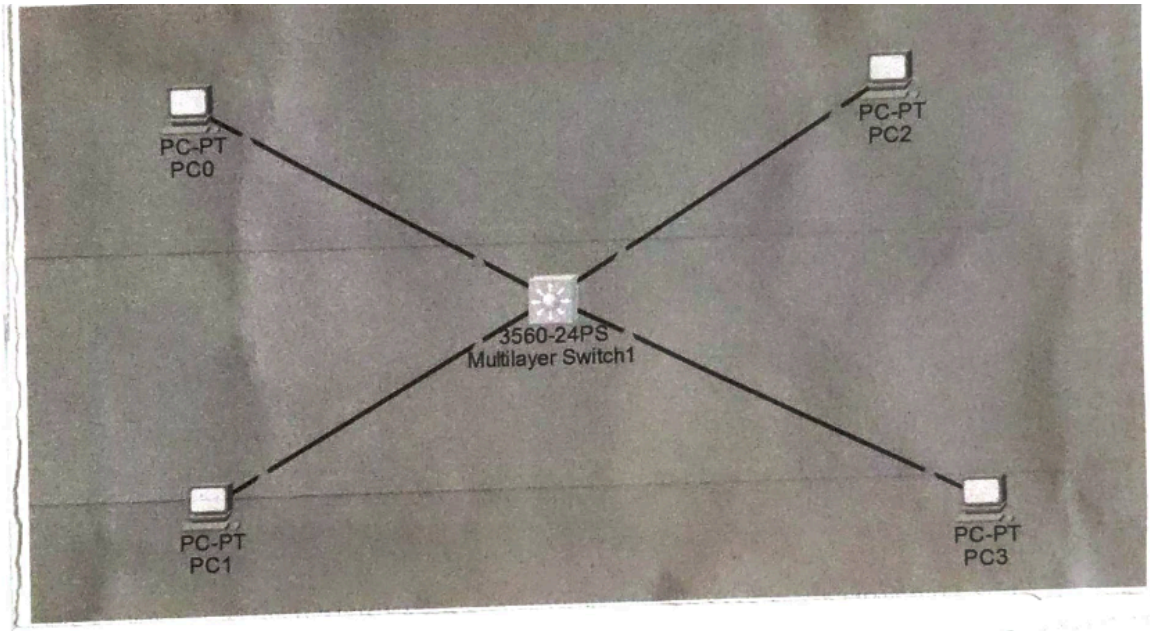
Experiment-4
LAN Configuration

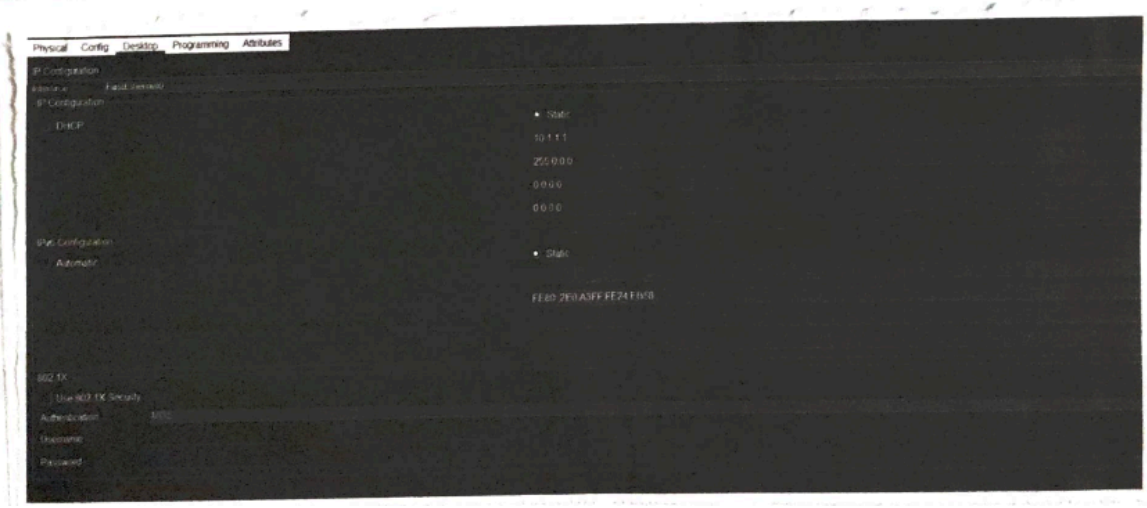
AIM:

To setup and configure a Local Area Network (LAN) using a Switch and Ethernet cables and verify connectivity between the connected computers.

PROCEDURE:

1. Take 4 computers, one network switch and 4 Ethernet cables.
2. Connect each computer to the Switch using Ethernet cables.
3. Assign IP addresses to the computers.
   - PC1 - 10.1.1.1
   - PC2 - 10.1.1.2
   - PC3 - 10.1.1.3
   - PC4 - 10.1.1.4
4. Assign Subnet mask:
   - 255.0.0.0
5. configure the switch with IP address 10.1.1.5.
6. Open Command prompt.
7. Use the ping command between the PCs.
8. Verify Successful connectivity.
9. Share a folder on one computer.
10. Access the shared folder from another computer


PC-PT
PC0
PC-PT
PC2
3560-24PS
Multilayer Switch1
PC-PT
PC1
PC-PT
PC3


Physical
Config
Desktop
Programming
Attributes
IP Configuration
DHCP
Static
10.1.1.1
255.0.0.0
0.0.0.0
0.0.0.0
IPv6 Configuration
Automatic
Static
802.1X
Use 802.1X Security
Authentication
Username
Password

These IP configurations and the LAN setup procedure are given in the manual.

Sample Output:

PC1> Ping 10.1.1.2

Reply from 10.1.1.2:

Bytes = 32 Time<1ms TTL=128

Packets: Sent = 4

Received = 4

Lost = 0

0 % packet loss

Shared folder:

\\10.1.1.1\SharedFolder

Access successful.

9/10

Result:

Thus, a LAN was successfully configured using a switch and Ethernet cables, and connectivity and file sharing b/w the systems were verified.